Fourier Series and the Circuit Design of a 5 V USB Charger

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***Abstract*—This report develops the Fourier-series and Fourier-transform analysis of a full-wave rectified waveform, then connects that analysis to the practical stages of a 5 V charger: voltage conversion, rectification, smoothing, regulation, and the USB output interface. Circuit explanations are expanded for intuitive understanding, while numerical values are distinguished from reference examples and unverified assumptions.**

1 Periodic Function

Let

$$x(t) = A_0 \left|\sin\left(2\pi f_0 t\right)\right| \tag{1.1}$$

1.1 Plot $x(t)$.

Solution. The plot of $x(t)$ is shown in Fig. 1.1.

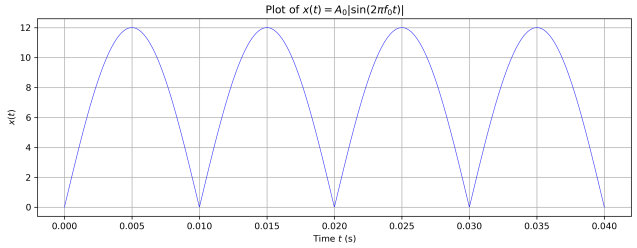


Fig. 1.1. Plot of $x(t) = A_0 \left|\sin(2\pi f_0 t)\right|$.

1.2 Show that $x(t)$ is periodic and find its period.

Solution. Note that,

$$x\left(t + \frac{1}{2f_0}\right) = A_0 \left|\sin\left(2\pi f_0 \left(t + \frac{1}{2f_0}\right)\right)\right| \tag{1.2}$$

$$= A_0 \left|\sin\left(2\pi f_0 t + \pi\right)\right| \tag{1.3}$$

$$= A_0 \left|-\sin\left(2\pi f_0 t\right)\right| \tag{1.4}$$

$$= A_0 \left|\sin\left(2\pi f_0 t\right)\right| \tag{1.5}$$

$$= x(t). \tag{1.6}$$

Hence, $x(t)$ is periodic with fundamental period

$$\boxed{T_0 = \frac{1}{2f_0}}. \tag{1.7}$$

2 Fourier Series

Consider $A_0 = 12$ and $f_0 = 50$ for all numerical calculations.

2.1 If

$$x(t) = \sum_{k=-\infty}^{\infty} c_k e^{j2\pi k f_0 t} \tag{2.1}$$

show that

$$c_k = f_0 \int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} x(t) e^{-\mathrm{j}2\pi k f_0 t}\, dt. \tag{2.2}$$

Solution. For some $n \in \mathbb{Z}$, multiplying both sides of (2.1) by $e^{-\mathrm{j}2\pi n f_0 t}$ gives

$$x(t) e^{-\mathrm{j}2\pi n f_0 t} = \sum_{k=-\infty}^{\infty} c_k e^{\mathrm{j}2\pi (k-n) f_0 t}. \tag{2.3}$$

Now, integrating both sides over the interval $\left[-\frac{1}{2f_0}, \frac{1}{2f_0}\right]$,

$$\begin{aligned} &\int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} x(t) e^{-\mathrm{j}2\pi n f_0 t}\, dt \\ &= \sum_{k=-\infty}^{\infty} c_k \int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} e^{\mathrm{j}2\pi (k-n) f_0 t}\, dt. \end{aligned} \tag{2.4}$$

The integral on the right can be evaluated using the orthogonality property of complex exponentials:

$$\begin{aligned} &\int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} e^{\mathrm{j}2\pi (k-n) f_0 t}\, dt \\ &= \begin{cases} \dfrac{1}{f_0}, & k = n, \\ 0, & k \neq n. \end{cases} \end{aligned} \tag{2.5}$$

Note that the integration interval has length $1/f_0$. Although the fundamental period of the given signal $x(t) = A_0 |\sin(2\pi f_0 t)|$ is $1/(2f_0)$, the chosen Fourier basis $e^{\mathrm{j}2\pi k f_0 t}$ has period $1/f_0$. Thus, the integration interval covers two fundamental periods of the signal and one period of the Fourier basis.
Consequently, all terms in the summation vanish except the term for which $k = n$. Therefore,

$$\begin{aligned} &\int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} x(t) e^{-\mathrm{j}2\pi n f_0 t}\, dt \\ &= \frac{c_n}{f_0}. \end{aligned} \tag{2.6}$$

Hence,

$$\boxed{c_n = f_0 \int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} x(t) e^{-\mathrm{j}2\pi n f_0 t}\, dt}. \tag{2.7}$$

2.2 Find c_k for (1.1).
Solution. Using (2.2),

$$c_n = f_0 \int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} A_0 |\sin(2\pi f_0 t)|\, e^{-\mathrm{j}2\pi n f_0 t}\, dt. \tag{2.8}$$

Using Euler's identity,

$$e^{-\mathrm{j}\theta} = \cos\theta - \mathrm{j}\sin\theta, \tag{2.9}$$

we obtain

$$\begin{aligned} c_n = f_0 A_0 &\int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} |\sin(2\pi f_0 t)| \cos(2\pi n f_0 t)\, dt \\ &- \mathrm{j} f_0 A_0 \int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} |\sin(2\pi f_0 t)| \sin(2\pi n f_0 t)\, dt. \end{aligned} \tag{2.10}$$

Since $x(t)$ is even and the sine term is odd, the imaginary part vanishes. Thus,

$$c_n = 2 f_0 A_0 \int_{0}^{\frac{1}{2f_0}} \sin(2\pi f_0 t) \cos(2\pi n f_0 t)\, dt. \tag{2.11}$$

Using the identity

$$2\sin A \cos B = \sin(A+B) + \sin(A-B), \tag{2.12}$$

we get

$$\begin{aligned} c_n = f_0 A_0 &\int_{0}^{\frac{1}{2f_0}} \sin(2\pi (n+1) f_0 t)\, dt \\ &+ f_0 A_0 \int_{0}^{\frac{1}{2f_0}} \sin(2\pi (1-n) f_0 t)\, dt. \end{aligned} \tag{2.13}$$

For $n \neq \pm 1$, evaluating the integrals gives

$$c_n = \frac{A_0}{2\pi}\left[\frac{1-(-1)^{n+1}}{n+1} + \frac{1-(-1)^{1-n}}{1-n}\right] = \frac{A_0(1+(-1)^n)}{\pi(1-n^2)}. \tag{2.14}$$

For $n = \pm 1$, direct evaluation of the original integral gives $c_n = 0$.
Therefore,

$$\boxed{c_n = \begin{cases} \dfrac{2A_0}{\pi(1-n^2)}, & n \text{ even}, \\ 0, & n \text{ odd}. \end{cases}} \tag{2.15}$$

Here, $n = 0$ is included in the even case, giving $c_0 = 2A_0/\pi$.

2.3 Verify (2.1) using Python.
Solution. The Python code `codes/2_3.py` verifies (2.1) by comparing the original signal with its Fourier-series reconstruction, as shown in Fig. 2.3.

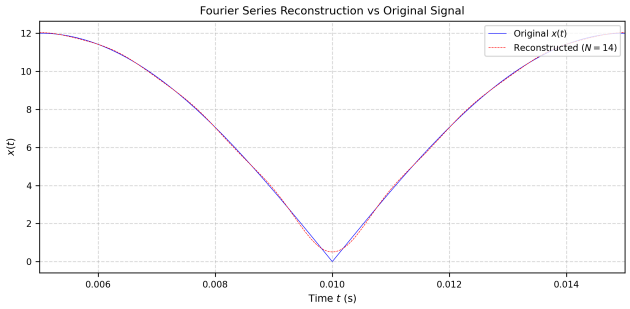


Fig. 2.3. Verification of the complex Fourier series.

2.4 Show that

$$x(t) = a_0 + \sum_{k=1}^{\infty}\left[a_k\cos(2\pi k f_0 t) + b_k\sin(2\pi k f_0 t)\right] \tag{2.16}$$

and obtain the formulae for a_k and b_k.
Solution. From (2.1),

$$x(t) = \sum_{k=-\infty}^{\infty} c_k e^{\mathrm{j}2\pi k f_0 t} \tag{2.17}$$

$$= c_0 + \sum_{k=1}^{\infty}\left[c_k e^{\mathrm{j}2\pi k f_0 t} + c_{-k}e^{-\mathrm{j}2\pi k f_0 t}\right]. \tag{2.18}$$

Using Euler's identities,

$$e^{\mathrm{j}\theta} = \cos\theta + \mathrm{j}\sin\theta, \tag{2.19}$$

$$e^{-\mathrm{j}\theta} = \cos\theta - \mathrm{j}\sin\theta, \tag{2.20}$$

we obtain

$$x(t) = c_0 + \sum_{k=1}^{\infty}\Big[(c_k + c_{-k})\cos(2\pi k f_0 t) + \mathrm{j}(c_k - c_{-k})\sin(2\pi k f_0 t)\Big]. \tag{2.21}$$

Comparing this with (2.16), we get

$$a_k = \begin{cases} c_0, & k = 0, \\ c_k + c_{-k}, & k > 0, \end{cases} \tag{2.22}$$

and

$$b_k = \mathrm{j}(c_k - c_{-k}), \qquad k > 0. \tag{2.23}$$

Using the coefficient formula (2.2), the corresponding integral expressions are

$$\begin{aligned} a_0 &= f_0\int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} x(t)\,dt, \\ a_k &= 2f_0\int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} x(t)\cos(2\pi k f_0 t)\,dt, \quad k > 0, \\ b_k &= 2f_0\int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} x(t)\sin(2\pi k f_0 t)\,dt, \quad k > 0. \end{aligned} \tag{2.24}$$

2.5 Find a_k and b_k for (1.1).
Solution. Since $x(t)$ is an even function,

$$x(-t) = x(t). \tag{2.25}$$

From the complex Fourier-series representa-

tion,

$$x(-t) = \sum_{k=-\infty}^{\infty} c_k e^{-\mathrm{j}2\pi k f_0 t} \tag{2.26}$$

$$= \sum_{k=-\infty}^{\infty} c_{-k} e^{\mathrm{j}2\pi k f_0 t}. \tag{2.27}$$

Comparing this with

$$x(t) = \sum_{k=-\infty}^{\infty} c_k e^{\mathrm{j}2\pi k f_0 t}, \tag{2.28}$$

we obtain

$$c_k = c_{-k}. \tag{2.29}$$

Therefore, from (2.22) and (2.23),

$$a_k = \begin{cases} \dfrac{2A_0}{\pi}, & k = 0, \\ \dfrac{4A_0}{\pi(1-k^2)}, & k > 0,\ k \text{ even}, \\ 0, & k > 0,\ k \text{ odd}, \end{cases} \tag{2.30}$$

$$b_k = 0, \qquad k > 0. \tag{2.31}$$

For $A_0 = 12$, these become

$$a_0 = \frac{24}{\pi}, \tag{2.32}$$

$$a_k = \begin{cases} \dfrac{48}{\pi(1-k^2)}, & k > 0,\ k \text{ even}, \\ 0, & k > 0,\ k \text{ odd}, \end{cases} \tag{2.33}$$

$$b_k = 0. \tag{2.34}$$

2.6 Verify (2.16) using Python.
Solution. The Python code `codes/2_6.py` verifies (2.16) by comparing the original signal with its real Fourier-series reconstruction, as shown in Fig. 2.6.

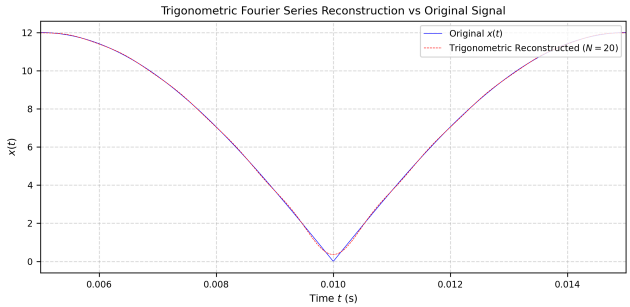


Fig. 2.6. Verification of the real Fourier series.

3 Fourier Transform

We use the Fourier-transform convention

$$G(f) = \int_{-\infty}^{\infty} g(t) e^{-\mathrm{j}2\pi f t}\, dt, \tag{3.1}$$

with inverse transform

$$g(t) = \int_{-\infty}^{\infty} G(f) e^{\mathrm{j}2\pi f t}\, df. \tag{3.2}$$

3.1 The unit impulse satisfies

$$\delta(t) = 0, \quad t \neq 0, \tag{3.3}$$

$$\int_{-\infty}^{\infty} \delta(t)\, dt = 1. \tag{3.4}$$

Solution. The Dirac delta is zero everywhere except at the origin, but its total integral is unity. More generally, it has the sifting property

$$\int_{-\infty}^{\infty} g(t)\delta(t - t_0)\, dt = g(t_0). \tag{3.5}$$

This property allows an impulse to select the value of a function at its location.

3.2 State the Fourier-transform pair for $g(t)$.
Solution. By definition, the Fourier transform of $g(t)$ is

$$G(f) = \int_{-\infty}^{\infty} g(t) e^{-\mathrm{j}2\pi f t}\, dt. \tag{3.6}$$

The corresponding inverse transform is

$$g(t) = \int_{-\infty}^{\infty} G(f) e^{\mathrm{j}2\pi f t}\, df. \tag{3.7}$$

Hence, we write $g(t) \overset{\mathcal{F}}{\longleftrightarrow} G(f)$.

3.3 Show that

$$g(t - t_0) \overset{\mathcal{F}}{\longleftrightarrow} G(f) e^{-\mathrm{j}2\pi f t_0}. \tag{3.8}$$

Solution. Let $y(t) = g(t - t_0)$. Its Fourier

transform is

$$Y(f) = \int_{-\infty}^{\infty} g(t - t_0) e^{-\mathrm{j}2\pi f t}\, dt. \tag{3.9}$$

Using the substitution $u = t - t_0$, so that $t = u + t_0$ and $dt = du$,

$$Y(f) = \int_{-\infty}^{\infty} g(u) e^{-\mathrm{j}2\pi f(u+t_0)}\, du \tag{3.10}$$

$$= e^{-\mathrm{j}2\pi f t_0} \int_{-\infty}^{\infty} g(u) e^{-\mathrm{j}2\pi f u}\, du \tag{3.11}$$

$$= G(f) e^{-\mathrm{j}2\pi f t_0}. \tag{3.12}$$

Therefore,

$$\boxed{g(t - t_0) \overset{\mathcal{F}}{\longleftrightarrow} G(f) e^{-\mathrm{j}2\pi f t_0}}. \tag{3.13}$$

3.4 Show that

$$G(t) \overset{\mathcal{F}}{\longleftrightarrow} g(-f). \tag{3.14}$$

Solution. From the inverse Fourier-transform relation,

$$g(-f) = \int_{-\infty}^{\infty} G(\nu) e^{-\mathrm{j}2\pi \nu f}\, d\nu. \tag{3.15}$$

This is precisely the Fourier transform of the function $G(t)$, evaluated at frequency f. Thus, by the duality property,

$$\boxed{G(t) \overset{\mathcal{F}}{\longleftrightarrow} g(-f)}. \tag{3.16}$$

3.5 Find the Fourier transform of $\delta(t)$.
Solution. Using the definition,

$$\mathcal{F}\{\delta(t)\} = \int_{-\infty}^{\infty} \delta(t) e^{-\mathrm{j}2\pi f t}\, dt. \tag{3.17}$$

By the sifting property, the integrand is evaluated at $t = 0$:

$$\boxed{\delta(t) \overset{\mathcal{F}}{\longleftrightarrow} 1}. \tag{3.18}$$

3.6 Find the Fourier transform of $e^{-\mathrm{j}2\pi f_0 t}$.
Solution. The standard frequency-shift impulse pair is

$$e^{\mathrm{j}2\pi f_0 t} \overset{\mathcal{F}}{\longleftrightarrow} \delta(f - f_0). \tag{3.19}$$

Replacing f_0 by $-f_0$ gives

$$\boxed{e^{-\mathrm{j}2\pi f_0 t} \overset{\mathcal{F}}{\longleftrightarrow} \delta(f + f_0)}. \tag{3.20}$$

3.7 Find the Fourier transform of $\cos(2\pi f_0 t)$.
Solution. Using Euler's identity,

$$\cos(2\pi f_0 t) = \frac{1}{2}\left(e^{\mathrm{j}2\pi f_0 t} + e^{-\mathrm{j}2\pi f_0 t}\right). \tag{3.21}$$

By linearity and the exponential transform pairs,

$$\boxed{\cos(2\pi f_0 t) \overset{\mathcal{F}}{\longleftrightarrow} \frac{1}{2}\left[\delta(f - f_0) + \delta(f + f_0)\right]}. \tag{3.22}$$

3.8 Find the Fourier transform of $x(t)$ and plot it. Verify using Python.
Solution. The signal is periodic with fundamental frequency $2f_0$. Its complex Fourier-series representation can therefore be written as

$$x(t) = \sum_{m=-\infty}^{\infty} d_m e^{\mathrm{j}2\pi(2mf_0)t}, \tag{3.23}$$

where, from the coefficients obtained in Section 2,

$$d_m = c_{2m} = \frac{2A_0}{\pi(1 - 4m^2)}. \tag{3.24}$$

The Fourier transform of a periodic signal is an impulse train in the frequency domain. Hence,

$$X(f) = \sum_{m=-\infty}^{\infty} d_m \delta(f - 2mf_0) \tag{3.25}$$

$$= \boxed{\sum_{m=-\infty}^{\infty} \frac{2A_0}{\pi(1 - 4m^2)} \delta(f - 2mf_0)}. \tag{3.26}$$

For $A_0 = 12$ and $f_0 = 50$ Hz,

$$X(f) = \sum_{m=-\infty}^{\infty} \frac{24}{\pi(1 - 4m^2)} \delta(f - 100m). \tag{3.27}$$

Python code plots the first few spectral lines and compares the analytical Fourier-series reconstruction with the original signal.

3.9 Show that $\operatorname{rect} t \overset{\mathcal{F}}{\longleftrightarrow} \operatorname{sinc} f$. Verify using Python.

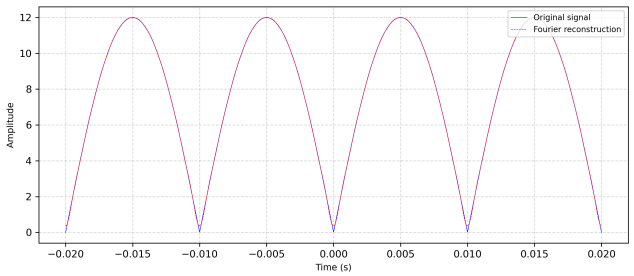


Fig. 3.8. Comparison with the Fourier reconstruction.

Solution. Using the normalized definitions

$$\operatorname{rect} t = \begin{cases} 1, & |t| < \frac{1}{2}, \\ 0, & |t| > \frac{1}{2}, \end{cases} \tag{3.28}$$

$$\operatorname{sinc} f = \frac{\sin(\pi f)}{\pi f}, \qquad \operatorname{sinc} 0 = 1, \tag{3.29}$$

the Fourier transform is

$$R(f) = \int_{-\infty}^{\infty} \operatorname{rect} t e^{-\mathrm{j}2\pi f t}\, dt \tag{3.30}$$

$$= \int_{-1/2}^{1/2} e^{-\mathrm{j}2\pi f t}\, dt \tag{3.31}$$

$$= \left[\frac{e^{-\mathrm{j}2\pi f t}}{-\mathrm{j}2\pi f}\right]_{-1/2}^{1/2} \tag{3.32}$$

$$= \frac{e^{-\mathrm{j}\pi f} - e^{\mathrm{j}\pi f}}{-\mathrm{j}2\pi f} \tag{3.33}$$

$$= \frac{\sin(\pi f)}{\pi f}. \tag{3.34}$$

Therefore,

$$\boxed{\operatorname{rect} t \overset{\mathcal{F}}{\longleftrightarrow} \operatorname{sinc} f}. \tag{3.35}$$

The following code numerically evaluates the transform and compares it with the analytical sinc function.

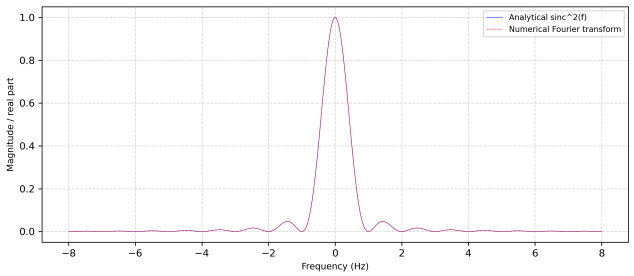


Fig. 3.9. Comparison with the Fourier transform.

3.10 Find the Fourier transform of $\operatorname{sinc} t$. Verify using Python.

Solution. By the duality property, since

$$\operatorname{rect} t \overset{\mathcal{F}}{\longleftrightarrow} \operatorname{sinc} f, \tag{3.36}$$

we obtain

$$\operatorname{sinc} t \overset{\mathcal{F}}{\longleftrightarrow} \operatorname{rect} -f. \tag{3.37}$$

The rectangular function is even, so $\operatorname{rect} -f = \operatorname{rect} f$. Hence,

$$\boxed{\operatorname{sinc} t \overset{\mathcal{F}}{\longleftrightarrow} \operatorname{rect} f}. \tag{3.38}$$

The following code numerically evaluates the transform over a finite time interval. Increasing the integration range and sampling density improves the approximation. At the discontinuities $f = \pm\frac{1}{2}$, the Fourier-transform value is conventionally taken as the midpoint value $1/2$.

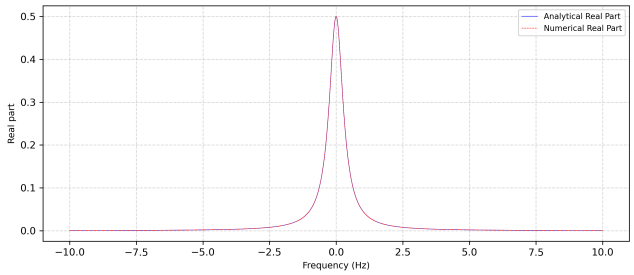


Fig. 3.10. Numerical verification of the Fourier transform of $\operatorname{sinc} t$.

4 Filter

4.1 Find $H(f)$ which transforms $x(t)$ to DC 5V.
4.2 Find $h(t)$.
4.3 Verify your result using through convolution.

5 Filter Design

5.1 Design a Butterworth filter for $H(f)$.
5.2 Design a Chebyschev filter for $H(f)$.
5.3 Design a circuit for your Butterworth filter.
5.4 Design a circuit for your Chebyschev filter.

TABLE 4
INPUT-STAGE VALUES USED IN THIS REPORT

Quantity	Symbol	Value
Primary RMS voltage	V_p	220 V
Mains frequency	f	50 Hz
Secondary RMS voltage (measured)	V_s	10.53 V
One silicon diode drop (assumed)	V_D	0.7 V
Bridge output ripple frequency	f_r	100 Hz
Reservoir capacitor in main draft	C	2200 μF

6 Circuit: From AC Input to Smoothed DC

Purpose and signal path

The circuit discussion follows the energy-conversion path up to the smoothing capacitor. The stages have different jobs: the transformer provides voltage reduction and isolation, the bridge rectifier makes the load polarity unidirectional, and the capacitor reduces the remaining periodic variation. The regulator and USB output are discussed in the following sections.

- **Input stage:** a 50 Hz AC source is the reference input.
- **Voltage conversion and isolation:** a step-down transformer provides a lower AC voltage and galvanic isolation.
- **Rectification:** a four-diode bridge converts both half-cycles into positive output pulses.
- **Smoothing:** a reservoir capacitor stores energy near each peak and supplies the load between peaks.
- **Regulation:** a separate regulator produces a more stable 5 V rail.
- **USB interface:** the port presents the correct electrical and charging-identification behaviour to a device.

Input quantities and notation

The calculations below use the measured secondary voltage from the input-stage analysis. This is kept distinct from the separate 12–0–12 V transformer example in the reference circuit report.

The symbols used throughout are:

- V_{rms}: root-mean-square voltage;
- V_{pk}: peak voltage;
- V_{avg}: average value over a cycle;
- $V_{r(pp)}$: peak-to-peak ripple voltage;
- I_L: load current; and
- f_r: frequency of the rectified ripple.

Mains waveform: RMS and peak values

A sinusoidal voltage is commonly specified by its RMS value because that value describes its equivalent heating effect in a resistive load. For a pure sinusoid, the peak value is larger than the RMS value by a factor of $\sqrt{2}$:

$$\boxed{V_{\mathrm{pk}} = \sqrt{2}\, V_{\mathrm{rms}}.} \tag{6.1}$$

For the nominal input stated in the reference calculation,

$$V_{p,\mathrm{pk}} = 220\sqrt{2} \tag{6.2}$$

$$\approx 311.1\ \mathrm{V}. \tag{6.3}$$

This value is included to explain the notation and the transformer ratio; it is not an instruction to handle or measure mains voltage.

Intuition. RMS voltage is similar to an average *effectiveness* rating rather than the highest instantaneous value. A 220 V RMS sine wave does not remain at 220 V; it swings between positive and negative peaks of approximately 311 V.

Step-down transformer

A transformer transfers energy between windings through a changing magnetic field. In an ideal transformer, the voltage ratio follows the turns ratio:

$$\boxed{\frac{V_s}{V_p} = \frac{N_s}{N_p}.} \tag{6.4}$$

Here, N_p and N_s denote the primary and secondary turns. Using the measured secondary RMS value:

$$\frac{N_p}{N_s} = \frac{V_p}{V_s} = \frac{220}{10.53} \tag{6.5}$$

$$\approx 20.9 : 1. \tag{6.6}$$

The corresponding secondary peak is

$$V_{s,\mathrm{pk}} = \sqrt{2}(10.53) \approx 14.89\,\mathrm{V}. \tag{6.7}$$

The transformer performs two conceptually separate functions:

- **Voltage conversion:** the secondary voltage is lower than the primary voltage.
- **Galvanic isolation:** energy is transferred magnetically, without a direct conductive connection between primary and secondary windings in an appropriately designed isolation transformer.

Contemporary analogy. A transformer is like a contactless gearbox for AC energy: it changes the voltage/current relationship through magnetic coupling, much as gears trade speed for torque. Unlike a simple resistor divider, it can transfer useful power without intentionally burning the voltage difference as heat.

The reference main-circuit report describes a centre-tapped 12–0–12 V transformer and reports an approximately 12 V RMS secondary waveform, corresponding to about 18 V peak. That is a different stated transformer condition from the measured 10.53 V RMS value used in this report. These values must not be combined in one calculation. The actual transformer label, winding connection, load condition, and measured secondary voltage must be confirmed before any design is finalised.

Full-wave bridge rectifier

A bridge rectifier uses four diodes arranged so that the current through the load has the same direction during both half-cycles of the AC input. During one half-cycle, one diagonal pair conducts; during the next half-cycle, the other pair conducts. At any instant, the conducting path contains two forward-biased diodes.

- Positive input half-cycle: the first diode pair carries current through the load.
- Negative input half-cycle: the second pair carries current through the load in the same load direction.
- Result: the negative half-cycle is flipped upward, producing a full-wave rectified waveform.

Contemporary analogy. Think of a one-way street system: although the source alternates direction, the bridge routes both directions of travel so that traffic through the load always moves the same way.

For a silicon bridge, a first-order estimate uses two diode drops:

$$\boxed{V_{o,\mathrm{pk}} \approx V_{s,\mathrm{pk}} - 2V_D.} \tag{6.8}$$

TABLE 4
CALCULATED WAVEFORM VALUES AT THE BRIDGE OUTPUT

Quantity	Symbol	Value
Secondary RMS voltage	V_s	10.53 V
Secondary peak voltage	$V_{s,\mathrm{pk}}$	14.89 V
Bridge peak voltage	$V_{o,\mathrm{pk}}$	13.49 V
Bridge average voltage	$V_{o,\mathrm{avg}}$	8.59 V
Bridge RMS voltage	$V_{o,\mathrm{rms}}$	9.54 V
Ripple frequency	f_r	100 Hz
Ripple period	T_r	10 ms

With $V_D = 0.7\,\mathrm{V}$ and the measured secondary peak:

$$V_{o,\mathrm{pk}} = 14.89 - 2(0.7) \tag{6.9}$$

$$\approx 13.49\,\mathrm{V}. \tag{6.10}$$

The ideal full-wave rectified sine has average value

$$\boxed{V_{o,\mathrm{avg}} = \frac{2V_{o,\mathrm{pk}}}{\pi}} \approx 8.59\,\mathrm{V}, \tag{6.11}$$

and RMS value

$$\boxed{V_{o,\mathrm{rms}} = \frac{V_{o,\mathrm{pk}}}{\sqrt{2}}} \approx 9.54\,\mathrm{V}. \tag{6.12}$$

These are idealised waveform calculations. Real diode forward voltage depends on current and temperature, and transformer regulation and winding resistance also affect the measured result.

Since both input half-cycles become positive pulses, the pulse frequency doubles:

$$\boxed{f_r = 2f = 100\,\mathrm{Hz}, \qquad T_r = \frac{1}{f_r} = 10\,\mathrm{ms}.} \tag{6.13}$$

The bridge output is therefore *pulsating DC*, not yet a steady 5 V supply. Its voltage falls substantially between successive peaks when a load is connected.

Diode selection considerations. The reference summary compares the 1N4007 (nominal 1 A average forward-current class and 1000 V repetitive peak reverse-voltage rating) with the 1N5408 (3 A average forward-current class). The correct diode or packaged bridge must be selected using the actual load current, surge current, reverse-voltage stress, temperature, and datasheet conditions. A high reverse-voltage rating alone does not establish suitability for the required current or thermal conditions.

Smoothing capacitor: reservoir action

A capacitor connected across the rectifier output acts as an energy reservoir. Near each rectified peak, the rectifier voltage exceeds the capacitor voltage and the capacitor is replenished. As the rectifier voltage falls below the capacitor voltage, the diodes stop conducting and the capacitor supplies the load until the next peak.

- **Charging interval:** the capacitor voltage rises towards the rectifier peak.
- **Discharging interval:** the capacitor releases stored energy into the load between peaks.
- **Ripple:** the repeated charge/discharge cycle creates a small periodic variation superimposed on the DC level.

- **Larger capacitance:** for the same load current and ripple frequency, a larger capacitor generally reduces the voltage drop between peaks.

Contemporary analogy. The capacitor behaves like a small water tank placed after an intermittent pump. The rectifier refills the tank at each pulse; the load draws water continuously. A larger tank reduces the change in water level between refills. The analogy describes energy buffering, not a literal equivalence between voltage and water pressure.

For a resistive load R_L, the ideal discharge between charging peaks is described by

$$\boxed{v_C(t) = V_{\max} e^{-(t-t_0)/(R_L C)}.} \tag{6.14}$$

The associated time constant is

$$\boxed{\tau = R_L C.} \tag{6.15}$$

A larger τ means the capacitor voltage decays more slowly during the interval between rectifier peaks.

For a full-wave rectifier and relatively small ripple, the peak-to-peak ripple can be estimated by

$$\boxed{V_{r(pp)} \approx \frac{I_L}{f_r C}.} \tag{6.16}$$

For $f_r = 100\,\text{Hz}$ and $C = 2200\,\mu\text{F}$:

$$V_{r(pp)} \approx \frac{I_L}{100(2200 \times 10^{-6})} \tag{6.17}$$

$$\approx 4.545 I_L \text{ V}, \tag{6.18}$$

where I_L is expressed in amperes. The approximate average capacitor voltage is

$$\boxed{V_{\text{DC}} \approx V_{\max} - \frac{V_{r(pp)}}{2}.} \tag{6.19}$$

With $V_{\max} \approx 13.49\,\text{V}$, this becomes

$$V_{\text{DC}} \approx 13.49 - 2.273 I_L \text{ V}. \tag{6.20}$$

This is an estimate, not a measured output. It assumes small ripple and nearly constant load current. The load current, capacitor tolerance, equivalent series resistance (ESR), transformer impedance, diode conduction interval, and temperature all affect the actual ripple.

The reference circuit report mentions a $100\,\mu\text{F}$ capacitor as its filter element and reports an approximately 18 V DC level in its particular simulation/measurement. The present main draft uses $2200\,\mu\text{F}$ in its calculations. Those are not interchangeable experimental records. The capacitor value must be chosen from the intended load current and allowed ripple, and its voltage and ripple-current ratings must be suitable for the actual circuit. The reported 18 V result belongs to the reference's 12 V RMS / 18 V peak example, not to the 10.53 V RMS calculation above.

Important limitation. A reservoir capacitor does not produce a precision DC output. It reduces the low-frequency ripple but does not guarantee a fixed voltage as the load changes. It also does not replace the regulator, current limiting, thermal design, or USB source-identification circuitry.

7 Regulator and Buck Converter

The regulator stage converts the smoothed, load-dependent DC rail into a controlled output. The two devices considered here are alternatives, not stages that must be connected in series. Both can provide a nominal 5 V rail, but their energy-conversion mechanisms and practical limitations are different.

7.1 IC 7805 linear regulator

The 7805 is a fixed positive linear regulator. Its internal reference and error amplifier monitor the output and control a series pass transistor. If the output tends to fall under increased load, the control circuit increases conduction; if the output tends to rise, it reduces conduction. This feedback action keeps the output near 5 V while the device remains within its operating and thermal limits.

- Fixed nominal output: $V_{\mathrm{OUT}} \approx 5\,\mathrm{V}$.
- Requires input headroom above 5 V; a useful first estimate is $V_{\mathrm{IN}} \gtrsim V_{\mathrm{OUT}} + V_{\mathrm{dropout}}$.
- The TI LM340/LM7805 family documentation gives about 2 V dropout at a stated 1 A test condition. The exact value depends on device, load, and temperature.
- The excess voltage is mainly dissipated as heat in the pass element.

The approximate power dissipation is

$$\boxed{P_D \approx (V_{\mathrm{IN}} - V_{\mathrm{OUT}})I_{\mathrm{OUT}}.} \tag{7.1}$$

For an illustrative 12 V input, 5 V output, and 0.5 A load:

$$P_D \approx (12 - 5)(0.5) = 3.5\,\mathrm{W}. \tag{7.2}$$

At 1 A under the same voltage conditions, the estimate rises to 7 W. Such dissipation can make a small package very hot unless the thermal path is designed appropriately.

Thermal intuition. A 7805 is like a pressure-reducing valve that discards the excess pressure. The output is controlled, but the unused voltage difference is not recovered; electrically, much of that difference becomes heat.

A simplified junction-temperature estimate is

$$\boxed{T_J \approx T_A + P_D R_{\theta JA},} \tag{7.3}$$

where T_A is ambient temperature and $R_{\theta JA}$ is the effective junction-to-ambient thermal resistance of the assembled device. A heatsink or larger copper area can improve heat removal, but the actual thermal resistance depends on mounting and airflow.

The 7805 family includes current limiting, thermal shutdown, and safe-operating-area protection. These are protective functions, not permission to operate continuously at the protection threshold. A family capability above 1.5 A is conditional on device variant and thermal design; it is not a guaranteed continuous current for every low-cost module.

Advantages

- Simple circuit and straightforward fixed 5 V output.
- No switching-frequency ripple generated by the regulator.
- Useful when the input-output voltage difference and load current are modest.

Limitations

- Efficiency becomes poor when the input is much higher than 5 V.
- Dissipation increases with both voltage difference and output current.
- Input must remain above the required headroom even at ripple valleys and under load.

7.2 LM2596 buck converter

The LM2596 is a fixed-frequency step-down switching regulator. The family includes fixed-output versions and an adjustable version. The TI datasheet specifies a 4.5–40 V operating input range, a 3 A output-current capability under stated conditions, and a 150 kHz oscillator. The ratings of a complete module also depend on its inductor, diode, capacitors, PCB layout, and cooling.

A buck converter does not continuously burn off the excess input voltage. Instead, an internal switch transfers energy in pulses to an inductor. The inductor stores energy in its magnetic field and continues

delivering current when the switch is off. A catch diode provides the current path during the off interval, while the output capacitor reduces the resulting voltage variation.

- **Switch ON:** the internal switch connects the input to the inductor path; inductor current rises and energy is stored.
- **Switch OFF:** the switch opens; inductor current continues through the catch diode and supplies the load.
- **Output filtering:** the capacitor supplies or absorbs the difference between inductor current and load current.
- **Feedback:** the controller senses the output and adjusts switching behaviour to keep the average output near the set value.

Contemporary analogy. The inductor acts like a flywheel: it resists abrupt changes in current and keeps energy moving during the switch-off interval. The capacitor acts like a buffer tank for voltage. Together they turn a train of switching pulses into a much steadier output.

For an ideal buck converter operating in continuous-conduction mode, the average output is

$$\boxed{V_{\mathrm{OUT}} = DV_{\mathrm{IN}}, \qquad D = \frac{V_{\mathrm{OUT}}}{V_{\mathrm{IN}}},} \tag{7.4}$$

where D is the duty ratio. For an ideal 12 V to 5 V conversion:

$$D \approx \frac{5}{12} = 0.417 = 41.7\%. \tag{7.5}$$

Real output differs because of switch and diode drops, winding resistance, ripple, control limits, and load conditions.

The efficiency is

$$\boxed{\eta = \frac{P_{\mathrm{OUT}}}{P_{\mathrm{IN}}}.} \tag{7.6}$$

For illustration only, a 5 V, 1 A output is 5 W. At an assumed 85% efficiency, the input power would be approximately $5/0.85 = 5.88$ W and the loss approximately 0.88 W. This is an example, not a guaranteed efficiency figure for every LM2596 module.

Advantages

- Usually more efficient than a linear regulator for a substantial step-down ratio.
- Lower heat dissipation for comparable output power in many operating conditions.
- Fixed and adjustable output variants are available.

Limitations

- Requires an inductor, catch diode, and suitable input and output capacitors.
- Switching ripple and electromagnetic interference (EMI) must be considered.
- The 3 A IC rating does not mean every inexpensive module can deliver 3 A continuously.
- Inductor saturation, diode current, capacitor ripple rating, PCB copper, and temperature constrain usable output.

Selection for this project. If the smoothed input is substantially above 5 V and the USB load current is meaningful, the LM2596 is generally the more practical candidate because it avoids the large linear loss predicted by (7.1). A 7805 remains useful as a simple educational reference or for a low-current rail when its headroom and thermal conditions are verified. The regulator must be selected from the actual DC input range, not from the transformer's nominal label alone.

8 USB Output Port

A USB connector is not merely a pair of 5 V and ground contacts. It is an interface with power, return, data or identification contacts, mechanical requirements, and rules that tell a connected device

TABLE 4
PRACTICAL COMPARISON OF THE TWO 5 V REGULATOR OPTIONS

Feature	**7805**	**LM2596**
Method	Linear	Switching buck
Output	Fixed 5 V	Fixed or adjustable
Main loss	Voltage drop as heat	Switching and component losses
External parts	Bypass capacitors; thermal path	Inductor, diode, capacitors
Ripple character	No switching ripple	Switching ripple present
Best fit	Small drop, modest current	Larger voltage reduction
Main design check	Dissipation and headroom	Current, ripple, layout, heat

how much current it may draw. A regulator can hold a 5 V rail, but it does not automatically make the port a compliant charger.

Main functions of the USB output

- **VBUS:** carries the positive supply, nominally 5 V for the basic USB charging modes discussed here.
- **GND:** provides the return path for current.
- **D+ and D-:** USB 2.0 data lines; in legacy charging arrangements, their electrical signature also helps a device identify the source type.
- **CC pins on USB Type-C:** advertise source current and establish connection orientation/role information.
- **USB-PD communication:** negotiates supported voltage/current profiles when a suitable Power Delivery controller and power path are present.

Contemporary analogy. VBUS is the delivery lane carrying energy. The identification pins are the signboard at the entrance that tells the device what kind of lane it has entered and what current allowance is available. A powerful supply without the correct signboard may still be treated as a lower-capability source.

Voltage is not the same as current. The regulator establishes the output voltage. The connected device draws current according to its operating state and the source capability it detects. The charger must be able to maintain its voltage at that load, but it should not force a fixed current into every device. Current advertisement, current limiting, cable capability, and protection are separate design requirements.

Legacy USB downstream ports. A Standard Downstream Port (SDP) is a normal host/data port. The USB 2.0 reference limit is up to 500 mA after the applicable enumeration/configuration process; a USB 3.x SDP is up to 900 mA under its specification. These are not blanket promises that any device can immediately draw those currents from any USB socket. Host rules and device configuration apply.

Battery Charging Specification 1.2. BC1.2 distinguishes three common port types:

- **SDP (Standard Downstream Port):** supports USB data and follows the applicable USB host current rules.
- **CDP (Charging Downstream Port):** supports data and provides charging capability, with up to 1.5 A under the standard's conditions.
- **DCP (Dedicated Charging Port):** intended for charging rather than USB data; the D+ and D- lines have the BC1.2 DCP signature, commonly implemented as a short between the two lines. A compatible device can identify the port and draw up to 1.5 A within the specification.

A device must perform the relevant detection. Merely shorting pins or applying a voltage without following the applicable standard does not make a universal or compliant charging port.

**Host enumeration and configuration rules apply.*

TABLE 4
COMMON USB CHARGING PORT CATEGORIES

Port type	Data support	Reference current
USB 2.0 SDP	Yes	Up to 500 mA*
USB 3.x SDP	Yes	Up to 900 mA*
BC1.2 CDP	Yes	Up to 1.5 A
BC1.2 DCP	No data function	Up to 1.5 A

TABLE 4
LEGACY D+/D- VOLTAGE SIGNATURES RECORDED IN THE REFERENCE

D+	D-	Reported current
2.0 V	2.0 V	Approx. 0.5 A
2.0 V	2.7 V	Approx. 1.0 A
2.7 V	2.0 V	Approx. 2.1 A
2.7 V	2.7 V	Approx. 2.4 A

TABLE 4
WHAT THE REGULATOR DOES AND WHAT THE USB INTERFACE MUST DO

Regulator / power path	USB interface
Maintain the selected DC voltage	Identify the port type
Supply the required load current	Advertise or negotiate allowed current
Remain within thermal limits	Use suitable connector and cable
Provide appropriate electrical protection	Respect USB protocol requirements

Legacy proprietary D+/D- signatures. Some older charging ecosystems used voltage levels on D+ and D- to identify a source's current capability. The main draft's reference table records example Apple-style signatures. These values are legacy, proprietary arrangements and should not be treated as universal USB rules.

These are included to explain why older devices sometimes used different charging signatures. They are not a recommended implementation for a new universal charger. Device support varies, and the applicable source documentation must be checked.

USB Type-C current advertisement. Type-C uses Configuration Channel (CC) pins rather than relying only on the legacy D+/D- signatures. A compliant 5 V Type-C source can advertise default USB current, 1.5 A, or 3 A through the appropriate CC termination. The connected device reads that advertisement and draws only within the advertised capability. The cable and complete power path must also be rated for the current.

USB Power Delivery (USB-PD). USB-PD is a separate communication protocol. It allows a source and sink to negotiate supported voltage and current profiles. A USB-C connector connected to a fixed 5 V regulator does not automatically provide USB-PD. A suitable PD controller, compliant power-path design, cable, and negotiation process are required for higher-voltage profiles.

Output protection and verification. A practical low-voltage USB output needs a correctly rated current limit, short-circuit protection, overvoltage protection where appropriate, reverse-current considerations, suitable connector and cable ratings, and a stable output under load. The regulator's headline current rating alone does not establish the safe current capability of the complete port. Verification should use an appropriate electronic load and measurement equipment on an isolated low-voltage output; no USB output should be tested while the upstream mains side is exposed.

9 Engineering Review and Design Checklist

The preceding analysis shows that each stage has a distinct requirement. A correct calculation at one stage does not prove that the complete charger is suitable. The values must be checked as a connected system under realistic load and temperature conditions.

1) **Input definition:** record the actual secondary RMS voltage and frequency under the intended load. Do not mix the measured 10.53 V value with the separate 12–0–12 V reference example.
2) **Rectifier:** verify the bridge current and reverse voltage ratings, surge capability, diode losses, and temperature.
3) **Capacitor:** select capacitance from load current and allowable ripple; check voltage rating, polarity, ESR, and ripple-current rating.
4) **Regulator:** check the minimum input at ripple valleys, maximum input at peaks, output current, efficiency, dissipation, and thermal behaviour.
5) **USB port:** choose one supported charging standard and implement its identification/advertisement correctly.
6) **System verification:** check output voltage, ripple, temperature, and protection behaviour at no load and at the intended low-voltage test loads.

Source-data reconciliation. The reference circuit report describes a centre-tapped 12–0–12 V transformer, a 100 μF filter capacitor, and an approximately 18 V peak/filtered level in its reported setup. The input-stage summary supplied with this project instead uses a measured 10.53 V RMS secondary, calculates 14.89 V peak before the bridge, and 13.49 V peak after two diode drops. The main source file specifies 2200 μF. These are different data sets. This draft retains the 10.53 V measured case and 2200 μF as the main calculation, while explicitly identifying the 12–0–12 V / 100 μF values as reference-report data requiring confirmation.

The reference circuit's reported waveform labels also call the rectifier waveform a 12 V, 50 Hz rectified sinusoid in one place, while the results list an 18 V peak after transformer and rectifier. A full-wave bridge supplied at 50 Hz produces a 100 Hz ripple waveform. The 12 V and 18 V descriptions are therefore not sufficient by themselves to establish a consistent measured waveform. The calculation in this report uses the explicit RMS/peak equations and labels the resulting ripple frequency correctly.

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